MCS 320 Quiz 7 Friday 16 October 2026

Submit your Jupyter notebook with the answers to gradescope by 1:50pm.

1. Define a function `P` which returns $p_i(x_1, x_2, \ldots, x_n) = x_i x_1 - x_{i+1}$, for $i = 1, 2, \ldots, n-1$.
   The input to `P` is the index i and the list $x_1, x_2, \ldots, x_n$.
   Demonstrate that your function works for $n = 8$.

2. Let $P = (2, 3)$ be the point on the curve $x^2y^3 - 5x^3 - 7y^2 - 5 = 0$.
   Compute the tangent line to the curve at P. What is the slope of this tangent line?